REQUEST FOR PROPOSAL – RFP

Maritime Transformation Project

Unified Maritime Digital services Platform

July 2026

Release Date: **July 2026**

1. DEFINITIONS

- **"MOEI"** means the Ministry of Energy and Infrastructure, the contracting authority for this RFP and hereinafter referred to as the **Buyer**.
- **"Supplier"** means the business, company, or other legal entity that submits a Proposal in response to this RFP. The Supplier may also be referred to as the **"Bidder."**
- **"Invitee"** means the business, company, or other legal entity that is invited by MOEI to participate in this RFP, along with the official invitation letter.
- **"Assets/Sites"** means the Ministry of Energy and Infrastructure offices or designated sites.
- **"Proposal"** means the official document submitted by a Supplier in response to this RFP.
- **"Services"** means the services forming the scope of work of this RFP, including (but not limited to) design, development, deployment, integration, hosting, training, and operational support of the platforms.
- **"RFP"** means this Request for Proposal, including all attachments, annexures, and referenced documents.
- **"Submission Deadline"** means the final deadline for submission of Proposals in accordance with the instructions provided in this RFP.
- **"Working Day"** means any day other than Friday, Saturday, or official public holidays in the United Arab Emirates.
- **"Policy"** means the digital procurement policy of the Federal Government of the United Arab Emirates.

2. PURPOSE

The **Ministry of Energy and Infrastructure (MOEI)** invites proposals from qualified vendors to deliver a comprehensive solution for **the Maritime Transformation Project** in accordance with this RFP.

The objective of this procurement is to secure a partner with the proven capability to design and implement an **innovative, flexible, scalable, AI driven and user-centric solution** that will drive operational excellence, strengthen institutional efficiency, and directly support the Ministry's national mandate. Vendors are expected to demonstrate a **deep understanding of MOEI's strategic priorities and regulatory framework**, present practical and measurable outcomes, and commit to delivering sustainable value through long-term support, knowledge transfer, and continuous improvement.

The system must leverage a configurable, microservices based, low code/no code, AI ready architecture to enable MOEI to modify workflows, rules, forms, fees, and eligibility logic without vendor dependency. It must ensure seamless integration with UAE PASS, GISIS/IMO, MOHRE, federal identity systems, Emirates Classification entities, port authorities, inspection databases and any other relevant entity that will be defined later during the detailed scoping phase of this project.

3. BACKGROUND

3.1 Ministry Overview

The **Ministry of Energy and Infrastructure (MOEI)** is a federal authority of the United Arab Emirates mandated to oversee and advance the development of the energy, infrastructure, housing, and transport sectors. The Ministry plays a central role in formulating national policies, implementing strategic initiatives, and managing projects that strengthen the UAE's economic growth and infrastructure readiness.

MOEI is dedicated to enhancing efficiency, promoting innovation, and ensuring the delivery of world-class services that meet the needs of citizens, residents, and future generations. Through effective governance and collaboration with stakeholders, the Ministry continues to build and maintain integrated systems that support the UAE's long-term vision and global competitiveness.

3.2 Vision

MOEI aspires to achieve sustainable global leadership in the fields of energy, water, infrastructure, housing, and transportation.

3.3 Mission

MOEI is committed to organizing, developing, and enhancing competitiveness across energy, mining, water resources, land and sea transportation, roads, utilities, housing, construction, and investment sustainability. The Ministry ensures the optimal use of partnerships, technology, and advanced sciences, while adopting global innovative solutions to improve the quality of life for society.

3.4 Core Values

MOEI's work is guided by the following core values: Role Model, Customer Focus, Innovation & Excellence, Teamwork, Trust, Agility & Primitiveness, and Competitiveness.

3.5 Maritime Sector Mandate

The Ministry of Energy and Infrastructure serves as the UAE's federal maritime administration and flag state authority. MOEI is responsible for the full spectrum of maritime regulatory governance, including vessel registration and certification, navigation licensing, seafarer licensing and competency verification, port oversight, marine facility regulation, and maritime safety. MOEI represents the UAE at the International Maritime Organization (IMO) and ensures national compliance with international maritime conventions including SOLAS, STCW, MLC, and MARPOL. It also focuses on the global and regional competitiveness of the UAE's maritime sector to ensure the highest quality of maritime services and the continuity of supply chains.

The Six Pillars of UAE Federal Maritime Administration

MOEI's maritime mandate operates across six fundamental functions that collectively define the full scope of this digital transformation:

- Regulatory Role: Regulating all means and modes of maritime transport and its personnel — including seafarers, commercial ships, ports, and supporting marine industries and services.
- Oversight Role: Maritime safety, environmental protection, maritime security, safety of navigation, and coastal services.
- Developmental Role: Comprehensive planning for the enablers and activities of maritime transport and ports.
- Administrative & Services Provider Role: Marine services delivery, maritime sector management and governance, quality management, information preservation, electronic archiving, and government and international relations.
- Advisory Role: Providing specialized maritime studies and consultations to UAE federal institutions.
- Complementary Economic Role: Enhancing the economic efficiency of maritime transport activities, boosting competitiveness of national ships and ports, and increasing their contribution to national income and GDP.

The platform must serve and enable all six functions — not only frontline service delivery, but also MOEI's supervisory, developmental, advisory, and economic mandate. It must equally support the strategic diplomatic objectives of the UAE Permanent Representative to the International

Maritime Organization (IMO), ensuring that the platform serves as a tool for global maritime engagement and flag state obligations, not only domestic service delivery.

3.6 Current State Challenges

The current delivery model for maritime services is characterized by the following systemic challenges:

A. Fragmented Platforms: More than 80 maritime services are distributed across disconnected systems with no unified platform, no single master data record, and no cross-service data sharing. Each domain operates in isolation.

B. Repeated Customer Effort: Applicants re-submit identical documents and information across multiple service applications. Vessel owners, seafarers, training centers, and marine companies experience repeated registration and data entry with no carryover between services.

C. Manual and Paper-Based Processes: Critical validations — insurance verification, classification checks, manning compliance, seafarer eligibility, and inspection recording — remain manual or semi-manual, creating bottlenecks and error risk.

D. No Unified Inspection Framework: Vessel, port, facility, and training center inspections each use separate methodologies and tools with no consolidated engine or shared compliance intelligence.

E. Limited National Oversight: The National Maritime Centre lacks a unified operational platform. Real-time visibility of vessel movements, compliance status, and risk patterns is not available in an integrated format. There is no unified platform for data linking and integration with relevant entities, nor a platform to provide data to decision-makers based on information from other sources that has been analyzed within the Center's operations

F. Inconsistent Customer Experience: Different service categories present entirely different user journeys, portals, and support models with no unified stakeholder experience.

G. Regulatory Enforcement Gaps: The absence of integrated data across all domains limits MOEI's ability to enforce compliance proactively, detect anomalies in real time, or generate intelligence for policy decisions.

H. Limited AI Adoption: Existing systems have no AI capabilities. The vast majority of decisions, validations, document reviews, and compliance checks are performed manually — a condition that must be fundamentally reversed in line with national AI directives. Focusing on developing services for individuals and the business sector by incorporating artificial intelligence technology

3.7 Strategic Alignment

This transformation initiative is directly aligned with the following national mandates and leadership directives:

- **UAE Zero Government Bureaucracy Program** — eliminating unnecessary steps, duplication, and procedural burden across all public services.
- **UAE Digital Government Strategy** — advancing seamless, data-driven, integrated digital services for all maritime stakeholders.
- **UAE Artificial Intelligence Strategy 2031** — establishing the UAE as a world leader in AI adoption across all government sectors.
- **HH Sheikh Mohammed bin Rashid Al Maktoum's Directive on Agentic AI** — a national mandate requiring 50% of UAE government services to operate through AI agents within 1 to 2 years, fundamentally transforming how government services are delivered.
- **UAE National Maritime Ambitions** — positioning the UAE as a globally recognized leader in maritime governance, safety, and digital innovation.
- **IMO Digital Maritime Agenda** — ensuring the UAE flag state meets and exceeds international expectations for data transparency, compliance reporting, and maritime safety.
- IMO Member State Audit Scheme (IMSAS) the platform must support IMSAS audit preparation and compliance monitoring, ensuring the UAE maintains comprehensive digital records, audit trails, and evidence of flag state obligation fulfilment required for IMSAS readiness.

"His Highness Sheikh Mohammed Bin Rashid announces 50% of UAE government services to run on AI agents in 2 years." — This directive is a binding strategic imperative for the Unified Maritime Digital Services Platform. The platform must be designed and delivered with AI-native agentic capabilities at its core, not as an afterthought. MOEI is committed to achieving and exceeding this target within the maritime services domain.

4. INFORMATION AVAILABLE

This RFP provides the Suppliers with the relevant required information. Further requests may be forwarded by email to the contact person listed in Section 21.

5. SCOPE OF WORK

5.1 Project Overview

The Ministry of Energy and Infrastructure (MOEI), Maritime Sector, is undertaking a comprehensive national digital transformation to consolidate, modernize, and unify all maritime regulatory, operational, and stakeholder-facing services into a single integrated AI-native platform. The Unified Maritime Digital Services Platform will serve as the authoritative digital backbone for the entire UAE maritime ecosystem — spanning ships, seafarers, academies, ports, marine facilities, inspections, legislation, and national maritime oversight.

This platform is not a conventional digitization project. It is an AI-first transformation — one in which autonomous AI agents will perform, support, and accelerate maritime service delivery across all seven business domains. The platform must be built from day one to place AI at the center of every service workflow, not as an optional add-on.

The platform must be built on a modern microservices-based, low-code/no-code, AI-native architecture — enabling MOEI to configure, adapt, and evolve all services and AI behaviors independently without vendor dependency.

5.2 Objectives

The platform must achieve the following measurable objectives:

- **O1 — Unified Digital Ecosystem:** Consolidate all 80+ maritime services across all 7 domains into a single platform with a unified data architecture, eliminating information silos and enabling a single source of truth for all maritime records.

- **O2 — Vendor Independence:** Empower MOEI business teams to independently configure forms, workflows, rules, fees, eligibility logic, AI agent behaviors, and service behavior without relying on the vendor for routine changes.

- **O3 — AI-Native Operations & Agentic Service Delivery:** Design and deliver the platform as an AI-first system where AI is embedded across all service domains. A minimum of 80% of the platform's maritime services must be AI-assisted within 24 months of go-live.

- **O4 — Smart Inspection:** Transform all inspection and audit activities into AI-powered operations where agents pre-assess risk, guide inspectors in real time, analyze evidence, detect anomalies, generate reports, and trigger enforcement actions autonomously — establishing the UAE as a global benchmark for AI-enabled maritime safety enforcement.

- **O5 — Superior Customer Experience:** Deliver a mobile-first, omnichannel experience with AI-assisted application journeys, personalized dashboards, real-time tracking, proactive AI-generated notifications, and zero-duplication experiences for all stakeholder categories.

- **O6 — Regulatory Compliance & Enforcement:** Strengthen MOEI's enforcement capabilities through AI-supported compliance monitoring, real-time alerts, cross-domain data linkage, predictive compliance monitoring, and alignment with applicable international conventions including SOLAS, STCW, MLC, and MARPOL.

- **O7 — National Maritime Intelligence:** Provide the National Maritime Centre with a real-time AI-synthesized operational picture of the UAE maritime domain — vessels, seafarers, ports, facilities, risks, and incidents — in a unified intelligence dashboard.

- **O8 — Bureaucracy Reduction:** Achieve measurable reductions in processing times, procedural steps, and customer effort. Target: Significant and measurable reduction in processing times, procedural steps, and customer effort across all maritime services.

5.3 Business Domains & Scope of Services

The platform must cover all seven maritime business domains defined by MOEI. Each domain must be fully digitized, AI-enabled, and integrated within the unified platform.

Domain 1 — Ships

This domain governs the full legal and regulatory lifecycle of all vessels operating under the UAE Flag or within UAE waters. It is the foundational domain upon which all other maritime services depend.

Scope of services:

- Vessel Registration: Provisional, permanent, amendments, and deletion from the UAE registry

- Vessel Certificates: Issuance, renewal, amendment, extension, and cancellation of CLC, CSR, exemption, ownership, and all statutory flag-state certificates
- Navigation License: Issuance, renewal, amendment, and suspension with automated dependency checks
- Foreign Vessel Permits: Risk-based evaluation and issuance for non-UAE-flagged vessels
- Vessel No Objection Certificates (NOC): Digitized submission, AI-assisted assessment, and automated restriction enforcement
- Vessel Transactions: Ownership transfer, mortgage registration and cancellation, sale contract attestation
- Vessel Master Record: Single authoritative vessel profile shared across all domains and services

AI requirements:

- AI agent for end-to-end processing of vessel registration renewals and certificate renewals — zero manual intervention for compliant vessels
- Automated AI validation of all submitted vessel documents against IMO standards and UAE regulations
- AI-generated compliance score for each vessel, updated in real time as data changes
- Predictive alerts for upcoming certificate expirations, enabling proactive renewal before lapses

Domain 2 — Seafarer's Affairs Administration & MET (Maritime Education and Training) Oversight

This domain governs the issuance of certificates, verification of professional competencies, and maintenance of seafarers' official service records. Additionally, it encompasses the approval, monitoring, and quality standards auditing of Maritime Education and Training (MET) institutions and their respective programs.

Scope of services:

- **Seafarer Certification & Endorsements**: The issuance, revalidation (renewal), upgrade, and suspension/revocation of Certificates of Competency (CoC), Certificates of Proficiency (CoP), and Flag State Endorsements, featuring automated compliance verification against STCW regulations.
- **Verification of Approved Seagoing Service**: Systematic validation of recorded sea service against official crew lists, discharges, and ship master/company logs via digitally verifiable service confirmations.

- **Statutory Certification Management**: Continuous tracking and registry maintenance of CoCs and CoPs, including automated alerts for revalidation deadlines and regulatory expiry dates.
- **MET Program Approval & Course Accreditation**: The formal evaluation, approval, and structured lifecycle management of maritime training programs to ensure strict alignment with STCW Code tables of competence.
- **MET Institution Accreditation & Quality Audit Integration**: The comprehensive approval lifecycle of maritime academies, integrated with periodic regulatory inspections and Quality Standards System (QSS) audits.
- Seafarer Master Profile: A unified, secure digital seafarer identity serving as a single source of truth across all administrative and regulatory services.
- Ship management Company, & Crewing Agency Integration: Secure, real-time portal access providing shipowners, managing companies, and crewing agencies with instant verification of license validity and STCW compliance.

Technical requirements: Specifications: Integrated Seafarer Registry

1. Unified Seafarer Database

Establish a centralized Seafarer Database utilizing a unique Seafarer Identification Number (SID) as the primary key. This registry must comprehensively capture and cross-reference all maritime personnel records, specifically targeting:

- Seafarers' Discharge Books (SDB): Full continuous discharge certificate data for all holders.
- Certificates of Competency (CoC): Comprehensive tracking of all active, suspended, or expired national CoCs.
- Maritime Training Records: Live tracking of all trainees and candidates successfully completing approved STCW courses across all recognized maritime training institutions.

2. Foreign CoC and Flag Endorsement Ledger

Maintain a distinct, dedicated ledger within the database for foreign-certified seafarers serving onboard UAE-flagged vessels. This ledger must track:

- Recognitions and Endorsements: Certificates of Receipt of Application (CRA) and formal Flag State Endorsements (under STCW Regulation I/10).
- Seaman Cards: Issuance and validity data for foreign crew members.

3. Vessel Linkage and Compliance Integration

The database architecture must provide dynamic, real-time data links between the seafarer registry, individual foreign ledger profiles, and the active fleet management modules, explicitly integrating:

- Minimum Safe Manning Documents (MSMD): Automated cross-referencing to ensure current vessel manning levels comply with the ship's specific MSMD requirements.

- Digital Crew Lists (IMO FAL Form 5): Direct synchronization with active crew lists to verify the credentials, medical certificates, and watchkeeping capacities of all crew members currently signed on UAE-flagged vessels.

In addition, we need to have a CoC verification webpage open and available to the public to verify the authenticity of our CoCs.

AI requirements:

- Automated Statutory Revalidation Agent: An autonomous processing agent for the streamlined revalidation of seafarer certifications, executing zero-touch processing for standard applications where all statutory requirements (medical fitness, approved sea service, or updated training) are fully met.
- STCW Gap Analysis & Training Matrices: AI-driven gap analysis to identify deficiencies in a seafarer's training portfolio relative to higher grades of certification or updated STCW amendments, automatically mapping required accredited courses.
- Cross-Validation of Authenticity: Automated crossmatching of submitted seafarer credentials against the active accreditation status and approved course listings of the issuing MET institution.
- Fraud Detection & Document Integrity: Advanced document intelligence models are designed to detect fraudulent, altered, or tampered certificates, ensuring the security of the administration's registry.

Domain 3 — Legislations, Regulations & Circulars

This domain manages the creation, publication, version control, and distribution of all maritime legislation, regulatory instruments, and official circulars issued by MOEI.

Scope of services:

- Legislative Repository: Centralized searchable digital library of all UAE maritime laws, ministerial decisions, and international conventions
- Circular & Notice Management: Issuance, versioning, distribution, and acknowledgement tracking
- Regulatory Change Management: Drafting, review, approval, and publishing workflows with impact assessment
- Compliance Linkage: Automated linking of regulations to specific vessel types, seafarer categories, and company activities
- Stakeholder Notification: Automated alerts to affected stakeholders on regulatory updates
- Public Portal: Open access to current maritime legislation, circulars, and regulatory guidance
- IMO Instrument Tracking: Monitoring and publication of IMO conventions and resolutions

AI requirements:

- AI agent for automated analysis of new IMO circulars and conventions, mapping their impact on the existing UAE regulations and affected services
- AI-powered regulatory Q&A assistant enabling stakeholders and MOEI staff to query legislation in natural language
- Automated detection of regulatory conflicts or gaps when new instruments are published
- AI-generated plain-language summaries of technical regulatory changes for stakeholder communication

Domain 4 — National Maritime Centre (NMC)

The National Maritime Centre functions as the UAE's federal maritime control tower — responsible for national-level maritime oversight, real-time monitoring, compliance intelligence, incident management, and strategic decision support.

Capabilities required:

- Real-Time Vessel Monitoring: AIS and LRIT integration for live vessel position, movement history, and port tracking
- Unified Compliance Dashboard: Real-time status of all UAE-flagged vessels, seafarers, companies, and training institutions
- Risk Intelligence Engine: AI-powered risk scoring for vessels, companies, and maritime operations
- Incident & Alert Management: Automated detection and escalation of safety incidents and regulatory violations
- Maritime Traffic Analytics: Historical and predictive analysis of vessel traffic patterns
- IMO GISIS Integration: Automated flag state data reporting
- Certificate Restriction Triggers: Automated suspension based on NMC-detected compliance failures
- Strategic Leadership Dashboards: Executive-level UAE maritime performance views
- Search and Rescue (SAR) Support: Integration with SAR coordination authorities

AI requirements:

- Autonomous AI agent continuously monitoring all vessel movements and flagging behavioural anomalies — unusual routes, suspicious loitering, AIS signal manipulation — without human initiation
- AI predictive models forecasting maritime incidents and safety events based on weather, vessel condition, route history, and crew records

- AI-generated daily and weekly maritime situation reports delivered automatically to NMC leadership
- Natural language interface for NMC operators to query the maritime domain ("show me all vessels overdue for inspection entering UAE waters today")

Domain 5 — Smart Inspection & Audit

The Smart Inspection & Audit domain transforms MOEI's enforcement function from a manual, paper-based activity into a fully AI-augmented operation — where artificial intelligence acts as an active participant in every stage of the inspection lifecycle: before, during, and after the physical inspection. AI agents will pre-assess risk, guide inspectors in real time, analyze captured evidence, generate findings reports, trigger corrective actions, and escalate critical issues — autonomously and continuously.

5.3.5.1 Pre-Inspection AI Intelligence

Before any inspector boards a vessel, enters a port, or visits a facility, AI agents must have already completed a comprehensive risk assessment and briefing:

- AI Pre-Arrival Risk Scoring: For every vessel entering UAE waters, an AI agent automatically calculates a composite risk score based on vessel age, flag state history, classification society, last inspection outcome, certificate status, crew records, deficiency history, route origin, cargo type, and global PSC detention records from Paris MOU, Tokyo MOU, and other regional agreements.
- Automated Inspection Targeting: AI agents autonomously select and schedule high-risk vessels and entities for priority inspection, without manual triage by MOEI staff — deploying inspector resources to where risk is highest.
- AI Pre-Inspection Dossier: Before each inspection, AI automatically compiles a structured briefing for the inspector covering vessel history, known deficiencies, outstanding corrective actions, similar vessel patterns, and predicted areas of non-compliance based on historical data.
- Predictive Deficiency Forecasting: AI models trained on global inspection data predict the specific deficiency categories most likely to be found on a given vessel — enabling targeted, intelligence-led inspections rather than generic checklists.
- Document Pre-Screening: AI agents pre-validate all certificates and documents on the vessel record before the inspector arrives, flagging expired, inconsistent, or potentially fraudulent documents with supporting evidence.

5.3.5.2 During-Inspection AI Assistance

During the physical inspection, AI acts as a real-time intelligent partner for the inspector:

- **AI-Guided Mobile Interface**: The inspector mobile application is driven by AI — dynamically adjusting the inspection checklist in real time based on vessel type, risk score, prior findings, and answers given during the inspection.
- **Computer Vision Evidence Analysis**: When an inspector captures a photo or video, an AI vision model immediately analyzes the image for visible deficiencies, safety hazards, equipment conditions, and compliance indicators — surfacing findings the inspector may have missed.
- **Real-Time Regulatory Lookup:** Inspector asks the AI ("is this fire suppression system compliant with SOLAS Chapter II-2?") and receives an instant, cited regulatory answer within the mobile app.
- **Voice-to-Inspection Transcription**: Inspectors can dictate findings verbally; AI transcribes, categorizes, and maps spoken observations to the relevant inspection checklist items and regulatory references automatically.
- **Anomaly Flagging**: AI compares live inspection findings against patterns from thousands of previous inspections — flagging when a declared condition is statistically inconsistent with vessels of the same type, age, and flag.
- **Offline AI Capability**: All AI inspection assistance functions offline onboard vessels without internet connectivity, synchronizing when connectivity is restored.

5.3.5.3 Post-Inspection AI Processing

After the inspection is completed, AI agents handle the entire post-inspection workflow autonomously:

- **Automated Inspection Report Generation**: AI compiles all captured findings, photos, voice notes, and checklist responses into a structured, formatted inspection report — ready for inspector review and signature within minutes of inspection completion
- **AI Deficiency Classification & Severity Scoring**: Each finding is automatically classified by deficiency category, regulatory reference, and severity level — with AI recommending detention, conditional release, or operational restrictions based on the severity profile
- **AI-Assisted Corrective Action Preparation**: AI agents automatically draft deficiency notices, corrective action instructions, and regulatory citations for standard deficiency types — with a designated MOEI officer reviewing and approving issuance before the notice is sent to the vessel operator
- **AI-Assisted Enforcement Recommendations**: AI agents automatically identify and recommend service-level actions based on inspection outcomes — including Navigation

License suspension for detained vessels, NOC restriction for non-compliant entities, and company license review triggers. Recommendations with legal or operational consequence are routed to a designated MOEI officer for final approval before execution

- **Flash Report Generation**: For critical safety findings, AI immediately generates an executive flash report and routes it to the appropriate NMC operator, MOEI management, and relevant authorities within seconds of the finding being recorded
- **Corrective Action Monitoring**: AI agents continuously track the status of all open corrective actions, sending automated reminders, escalating overdue items, and closing actions when verified evidence of rectification is received
- **Cross-Inspection Pattern Learning**: After each inspection, AI updates its models with the new data — continuously improving risk scoring accuracy, deficiency prediction, and anomaly detection across the entire inspection portfolio

5.3.5.4 Smart Inspection Coverage

Smart Inspection AI capabilities apply across all inspection and audit types within MOEI's mandate:

- **Vessel Inspections**: Port state control, flag state, offshore vessels, foreign vessel pre-arrival, post-incident reinspection.

5.3.5.5 Smart Inspection KPIs

The following AI-driven inspection performance targets must be achieved within 18 months of go-live:

- 100% of vessel inspections to be supported by an AI pre-inspection dossier and risk score before inspector deployment
- Minimum 70% of standard post-inspection reports to be AI-generated without manual drafting by inspectors
- Minimum 80% of standard corrective action notices to be AI-drafted and prepared for officer review and approval within 30 minutes of inspection completion
- AI deficiency prediction accuracy to achieve minimum 65% correlation with actual findings within 12 months
- Average inspector report completion time to be reduced by minimum 50% versus current baseline
- 100% of inspection-triggered service restriction recommendations to be generated by AI and routed to a designated MOEI officer for approval within 1 hour of inspection completion

Domain 6 — Ports

This domain manages MOEI's regulatory oversight of UAE port facilities in accordance with the International Ship and Port Facility Security (ISPS) Code.

It covers the full lifecycle of the Port Facility Statement of Compliance — the official certificate confirming that a port facility meets international security requirements — issued in coordination with the Federal Authority for Identity, Citizenship, Customs and Ports Security (ICP).

Scope of services:

- **Issue Port Facility Statement of Compliance**: End-to-end digital processing of new port compliance certificate applications, including submission, ICP integration for review and approval, fee payment, and digital certificate issuance
- **Annual Endorsement of Port Statement of Compliance**: Annual renewal and endorsement of existing port compliance certificates, with automated proactive notifications to port facility representatives ahead of endorsement deadlines — and automatic removal from the approved ports list if endorsement is not requested within the defined regulatory timeframe.
- **Amend Port Statement of Compliance**: Amendment of issued port compliance certificates to reflect changes in port facility data or vessel types, with ICP review automatically triggered when vessel type changes are involved

AI requirements:

- **AI-powered document intelligence** for automated reading and validation of all submitted port compliance attachments — including ownership documents, port sketches, and security officer documentation — verifying completeness, format, and data consistency before routing to ICP for review.
- Automated cross-referencing of submitted application data against ICP records, Ministry of Economy company registration data, and existing port compliance records — automatically detecting inconsistencies, duplicate submissions, or data conflicts before human review.
- **Automated processing of standard annual endorsement applications** where no vessel type changes are involved and all submitted documents are validated — automatically calculating fees, notifying the applicant, and issuing the endorsed certificate without manual MOEI intervention.
- Proactive compliance monitoring across the port compliance register, automatically identifying facilities approaching endorsement deadlines or at risk of removal from the approved ports list and generating targeted alerts to both facility representatives and MOEI officers at configurable intervals.

Domain 7 — Marine Facilities & Companies

This domain manages MOEI's regulatory accreditation and compliance oversight of specialized maritime service companies operating in the UAE. It covers the full lifecycle of the Annual Accreditation Certificate — the official MOEI certification confirming that a company is authorized to provide specialized maritime services on vessels operating under UAE jurisdiction.

The specialized service categories currently covered under this domain include magnetic compass correction and calibration, lifesaving equipment inspection and maintenance, firefighting equipment inspection and maintenance, survey of small vessels not subject to international maritime conventions, vessel pest control, and towage certificate issuance.

Scope of services:

- **Issue Annual Accreditation for Specialized Companies**: End-to-end digital processing of new accreditation applications, including company profile submission, equipment declaration, licensing authority validation, physical inspection visits where required, multi-level MOEI review and approval, fee payment, and accreditation certificate issuance.
- **Renew Annual Accreditation for Specialized Companies**: Annual renewal of existing accreditation certificates, with automated proactive notifications ahead of expiry, licensing authority re-validation, inspection visit where required, and certificate reissuance.
- **Inspection Visit Management**: A structured digital workflow enabling MOEI inspection teams to schedule, conduct, record, and manage physical site visits to specialized companies as part of accreditation and renewal decisions — including configurable inspection checklists, corrective action tracking, and full inspection history maintenance.

AI requirements:

- AI-powered document validation automatically verifying all submitted accreditation attachments — including company profiles, equipment approvals, IACS certificates, and factory authorization letters — for completeness and consistency before routing to MOEI reviewers.
- Automated cross-referencing of submitted company data against Ministry of Economy trade license records and ICP identity data — instantly detecting expired licenses, mismatched company details, or ineligible applicants before the application enters the review queue.
    - Proactive expiry monitoring across all accredited specialized companies, automatically identifying companies approaching certificate expiry or carrying overdue corrective actions and generating escalation alerts to the responsible MOEI officer.

5.4 Cross-Domain Platform Requirements

5.4.1 Unified Data Architecture

- Single master records for vessels, seafarers, companies, ports, and facilities — shared and reused across all domains with zero duplication
- A unified data model enabling cross-domain data queries, compliance checks, and AI analytics
- Real-time data synchronization ensuring all domains reflect the latest status of shared records

5.4.2 Integration Ecosystem

- UAE PASS: OpenID Connect / OAuth 2.0 federated identity for all customer authentication
- GISIS/IMO: Automated reporting of flag state data including vessel registry, seafarers, and casualty statistics
- MOHRE: Seafarer employment record validation and labor compliance verification
- Emirates Classification entities: Certificate status and survey record APIs
- AIS/LRIT providers: Real-time vessel position data feeds
- UAE National Single Window and Federal Identity Authority: as required by UAE Digital Government standards
- Open APIs: Published REST/SOAP API layer enabling future integrations

5.4.3 Customer Experience & Accessibility

- Mobile-first, omnichannel design — web browser, iOS, and Android with consistent experience
- Personalized stakeholder dashboards for each customer category
- Real-time application status tracking with proactive AI-generated notifications
- Arabic and English interface throughout with right-to-left layout support
- Compliance with TDRA Digital Government Standards

5.4.4 Configurability & Business Agility

- Low-code/no-code workflow and form designer enabling MOEI to create and modify service workflows independently
- Configurable AI agent behavior — MOEI business users can adjust AI decision thresholds, automation rules, and escalation conditions without vendor involvement
- Service versioning for regulatory transitions
- Full platform configurability including AI models, without developer dependency for routine changes

5.5 AI & Agentic Services — National Directive Implementation

5.5.1 Agentic AI Architecture

The platform must incorporate a dedicated AI Agent Layer — an orchestration framework enabling autonomous AI agents to perceive inputs, reason about context, execute multi-step tasks, make decisions, and take actions across all maritime service domains — with configurable levels of human oversight.

Mandatory agentic architecture requirements:

- **AI Agent Orchestration Platform**: A centralized agent management framework where MOEI can create, deploy, monitor, pause, and retire AI agents for any service workflow
- **Configurable Autonomy Levels**: Each AI agent must support configurable autonomy settings — from fully supervised (human approves every AI recommendation) to fully autonomous (AI executes and notifies) allowing MOEI to progressively increase automation as confidence grows.
- **Human-in-the-Loop Controls**: Every AI agent must have a clearly defined escalation path to a human operator for cases outside confidence thresholds, edge cases, or exceptional circumstances.
- **Agent Audit Trail**: Complete, immutable log of every AI agent decision, action, input, and output — providing full explainability and accountability for regulatory and legal purposes.
- Agent Performance Monitoring: Real-time dashboards showing each agent's decision volume, accuracy rates, escalation frequency, processing times, and business impact.
- **Multi-Agent Coordination:** Agents from different domains can collaborate and hand off tasks — for example, an inspection agent can instruct a licensing agent to suspend a navigation license based on inspection findings.

5.5.2 Mandatory AI Agents — Service Automation

The following AI agents must be delivered as part of the platform and must be operational within the timeframes specified, in line with the national directive:

Document Intelligence Agent

- Function: Reads, validates, and extracts structured data from all uploaded maritime documents — certificates, licenses, insurance policies, survey reports, Manning agreements
- Capability: OCR + AI understanding of document content, cross-reference against issuing authority databases and expected data formats

- Autonomy: Automatically approves documents meeting all validity criteria; flags exceptions for human review
- Target: Minimum of 80% of submitted documents processed without human intervention.

Vessel Compliance Agent

- Function: Continuously monitors the compliance status of every UAE-flagged vessel across all certificate types, inspection outcomes, insurance validity, and manning requirements
- Capability: Real-time compliance scoring, proactive expiry alerts, automated service restriction triggers
- Autonomy: Autonomously generates compliance reports, issues pre-expiry alerts, and initiates renewal workflows; escalates enforcement decisions to human reviewers
- Target: 100% of vessel compliance monitoring to be AI-automated

Service Processing Agent

- Function: Processes routine maritime service applications end-to-end — receiving submission, validating eligibility, checking compliance conditions, applying fees, issuing approvals or rejections
- Capability: Full workflow execution for standard cases meeting all defined eligibility criteria without human review
- Autonomy: Fully autonomous for standard renewal and low-complexity applications; supervised for new registrations, enforcement-linked applications, or cases with anomalies
- Target: 50% of eligible service applications to be fully processed by AI agents without human intervention

Customer Guidance Agent (Maritime Assistant)

- Function: An AI-powered conversational assistant available 24/7 across web and mobile channels, helping customers understand requirements, complete applications correctly, track their service status, and resolve common queries
- Capability: Natural language understanding in Arabic and English; context-aware guidance based on customer profile and service history; proactive notifications about upcoming renewals and compliance obligations
- **Autonomy**: Handles full customer interaction for standard inquiries; escalates to human MOEI staff for complex regulatory queries or complaints
- **Target**: 80% of customer inquiries resolved by AI assistants without human agent involvement.

Smart Inspection Agent

- **Function**: End-to-end AI support for all inspection activities — pre-inspection risk assessment and dossier generation, real-time guidance during inspection, post-inspection report generation, corrective action preparation, and enforcement action recommendations
- **Capability:** Computer vision analysis, predictive risk modeling, regulatory lookup, voice transcription, cross-inspection pattern recognition — as detailed in Domain 5
- **Autonomy:** Fully autonomous for pre-inspection preparation, evidence analysis, and report drafting. Corrective action notices are AI-drafted and submitted for officer approval before issuance. Any detention decision requires designated officer sign-off
- **Target**: 100% of inspections AI-supported; minimum 80% of inspection reports AI-drafted — timelines to be defined in the Smart Inspection domain go-live plan.

Regulatory Intelligence Agent

- **Function**: Monitors IMO publications, international maritime convention updates, and UAE regulatory developments — automatically analyzing their impact on existing platform services, configurations, and compliance rules
- **Capability**: Regulatory change impact mapping, automated update recommendations for affected service workflows, stakeholder notification generation
- **Autonomy**: Autonomous monitoring and impact analysis; recommends configuration changes to MOEI business teams for human approval before implementation
- **Target**: Operational from go-live; all new IMO circulars analyzed within 24 hours of publication.

National Maritime Intelligence Agent

- **Function**: Continuously synthesizes data from all platform domains and external data sources — AIS, LRIT, weather, port records, inspection history, global PSC databases — to generate a real-time maritime risk picture for the NMC
- **Capability**: Anomaly detection, predictive incident modeling, vessel behavioral analysis, automated situation reports
- **Autonomy**: Fully autonomous for monitoring, alerting, and reporting; human NMC operators review and act on AI-generated intelligence
- **Target**: Operational from go-live; real-time monitoring covering 100% of UAE-flagged and UAE-waters vessels

5.5.3 AI Governance & Ethics

All AI agents deployed within the platform must operate within a defined AI Governance Framework:

- Explainability: Every AI decision must be explainable — the platform must show which data, rules, and model outputs led to each AI-generated recommendation or action.
- Bias Monitoring: AI models must be regularly audited for systematic bias, particularly in risk scoring and inspection targeting, to ensure fair and non-discriminatory outcomes.
- Human Override: Any AI decision can be overridden by an authorized MOEI human reviewer at any time, with the override reason logged.
- Regulatory Compliance: AI agent behaviors must comply with UAE Personal Data Protection Law, UAE AI governance frameworks and guidelines as issued by the relevant UAE authorities, and any applicable IMO guidance on AI in maritime administration
- Model Versioning: All AI models must be version-controlled with the ability to roll back to a previous version if performance degrades.
- Continuous Learning: AI models must support supervised continuous learning — improving from new data while requiring MOEI approval before model updates are deployed to production.

5.5.4 AI Adoption Roadmap

Vendors must submit an AI Adoption Roadmap as part of their technical proposal, demonstrating a phased approach to deploying all AI capabilities across the seven business domains, with clear milestones, sequencing rationale, and proposed AI coverage targets at each phase.

5.6 Scope Exclusions

The following are explicitly excluded unless separately agreed in writing:

- Physical hardware procurement unless forming part of a proposed managed service
- Operational systems of individual ports that do not interface directly with MOEI regulatory services.
- Non-maritime services managed by other MOEI departments.
- Legacy system decommissioning beyond structured data migration to the new platform.
- Content creation or translation of existing legislation and circulars unless agreed as a separate workstream.
- Development of new maritime regulations or policies — the platform digitizes and enforces existing regulatory frameworks but does not create or amend them.
- Physical vessel surveys and inspections: the platform supports and records inspection outcomes but does not replace the physical presence of MOEI-certified surveyors and inspectors' onboard vessels.

5.7 Key Deliverables

- D0 — Data Readiness Assessment
- D1 — Platform Architecture & Design Documentation including AI Architecture
- D2 — Unified Maritime Digital Services Platform: all 7 domains and 80+ services
- D3 — AI Capabilities Suite: all AI capabilities and agents per Section 5.5
- D4 — Smart Inspection System: full AI-powered inspection capability per Domain 5
- D5 — Integration Layer: UAE PASS, GISIS/IMO, MOHRE, Classification entities, AIS/LRIT
- D6 — Mobile Applications: customer-facing and inspector-facing apps for iOS and Android
- D7 — NMC Intelligence Suite: real-time monitoring, AI risk engine, and strategic dashboards
- D8 — Data Migration: A data migration strategy covering source system analysis, data mapping, data cleansing requirements, migration approach, rollback plan, and validation criteria — to be approved by MOEI before migration commences, followed by structured execution with full validation and audit trail.
- D9 — UAT & Go-Live: Comprehensive testing strategy and execution covering all testing phases, user acceptance, defect resolution, and hypercare support — with detailed test plans and acceptance criteria to be agreed during project initiation.
- D10 — Training & Knowledge Transfer
- D11 — KPI & Performance Dashboards: real-time monitoring dashboards for all seven domains, executive leadership, and NMC
- D12 — AI Adoption Roadmap & Performance Reports: periodic AI performance reporting throughout the contract period

5.8 Functional Requirements

- End-to-end digital service workflows across all 7 domains with no manual handoffs.
- Configurable forms, workflows, rules, and fees fully manageable by MOEI without vendor involvement.
- Unified master records for vessels, seafarers, companies, ports, and facilities across all domains.
- Role-based access control for all user categories.
- Real-time application status tracking with AI-generated proactive notifications.
- Digital certificate generation, signing, and verification, meeting international standards.
- Smart Inspection system with AI pre-assessment, real-time guidance, and automated report generation.
- AI Agent Orchestration Platform with configurable autonomy levels and full audit trail
- UAE PASS federated authentication.
- Real-time data exchange with GISIS/IMO, MOHRE, AIS/LRIT, and classification society systems.
- NMC real-time monitoring with AI maritime intelligence.
- AI-enabled legislative repository with regulatory change management.
- Comprehensive audit trails and approval workflows across all services and AI decisions.

- Arabic and English multilingual interface with RTL support.
- 24/7 AI agent availability for customer guidance and service processing
- Domain KPI & Performance Dashboards: configurable dashboards for each of the seven business domains enabling MOEI to monitor and track business performance indicators in real time.

5.9 Non-Functional Requirements

- **Security:** UAE Cybersecurity Council standards; full encryption at rest and in transit.
- **Data Privacy:** UAE Personal Data Protection Law compliance.
- **Data Sovereignty:** All data and AI model training stored and processed within UAE jurisdiction.
- **Disaster Recovery:** RPO 4 hours; RTO 8 hours.
- **Mobile:** Native iOS and Android with offline AI capability for inspection workflows.
- **Accessibility:** Compliance with TDRA Digital Government Standards.
- **AI Explainability:** Every AI decision must produce a human-readable explanation of reasoning.
- **Low-Code/No-Code:** Full platform configuration manageable by non-technical MOEI staff without vendor involvement.

6. SERVICE LEVEL AGREEMENTS (SLAs)

Vendors must commit to the following minimum service levels:

6.1 Incident Response & Resolution

Metric	Target Response Time	Target Resolution Time	Support Hours
P1 – Critical Incident (system down, severe business impact)	Within 30 minutes	Within 2 hours	24/7 (or Business Hours – to be defined)
P2 – High Priority Incident (significant impact, workaround possible)	Within 2 hours	Within 4 hours	24/7
P3 – Medium Priority Incident (moderate impact, workaround available)	Within 6 hours	Within 24 hours	24/7

6.2 System Availability

Uptime: 99.9% (monthly, excluding scheduled maintenance)

6.3 Change Management

Metric	Target
Change Implementation Time	Within 24 hours (for standard, approved changes)
Out-of-Hours Requests	Must be supported and implemented as agreed

6.4 Performance SLAs

The platform must meet the following performance thresholds under peak production load, measured continuously from go-live:

Metric	Target	Measurement
Web page load time	Maximum 3 seconds	95th percentile under peak load
API response - standard queries	Maximum 500ms	95th percentile
API response - complex queries	Maximum 2 seconds	95th percentile
AI document verification	Maximum 30 seconds per document	95th percentile
AI inspection risk score	Maximum 60 seconds	95th percentile
AI conversational agent response	Maximum 5 seconds	95th percentile
NMC AIS vessel position refresh	Maximum 60-second latency	Rolling average
Digital certificate issuance	Maximum 10 seconds after approval	95th percentile
AI inspection report generation	Maximum 5 minutes after inspection close	95th percentile

6.5 Data, Security & Recovery SLAs

- Backup Completion: 100% of daily backups must complete; failure reported within 2 hours and remediated within 4 hours.
- Recovery Point Objective (RPO): Maximum 4 hours of data loss in any disaster recovery scenario.
- Recovery Time Objective (RTO): Maximum 8 hours for full platform restoration after a declared disaster.
- Security Patch - Critical (CVSS 9.0+): Applied within 24 hours of vendor notification.
- Security Patch - High (CVSS 7.0-8.9): Applied within 72 hours of vendor notification.
- Security Patch - Medium/Low: Applied within next scheduled maintenance window.
- Data Breach Notification: MOEI notified within 1 hour; preliminary impact assessment within 4 hours.
- Annual penetration test report delivered to MOEI within 10 working days of test completion.

6.6 AI-Specific SLAs

The following AI service levels apply in addition to the general SLAs above, reflecting the strategic importance of AI to this platform:

AI SLA Metric	Target	Review Frequency
Document Intelligence Agent Accuracy	Minimum 95% on standard maritime certificate validation	Monthly
AI Agent Availability	Each mandatory agent operational minimum 99.5% of scheduled hours	Monthly
Inspection Risk Scoring False Positive Rate	Maximum 15% on high-risk vessel targeting	Quarterly
AI Escalation Resolution Time	Human review of AI-escalated items within 4 business hours	Monthly
Model Performance Drift Alert	Automated alert when accuracy drops more than 5% from baseline	Continuous

AI Decision Explainability	Full explanation retrievable within 30 seconds of request	95th percentile
Agentic Service Rate - National Directive Progress • AI-Enabled Service Coverage: Minimum 80% of platform maritime services incorporating AI capabilities within 24 months of go-live. Progress tracked quarterly from go-live, reported in the Annual AI Adoption Report.	Quarterly progress report toward 50% target within 24 months	Quarterly

6.7 SLA Measurement, Reporting & Penalties

SLA performance will be continuously measured by an automated monitoring system agreed before go-live. Monthly compliance reports must be submitted by the vendor within 5 working days of each month end. MOEI reserves the right to independently verify all metrics through third-party monitoring tools at any time.

Vendors must include in their commercial proposal a financial penalty framework for SLA breaches covering availability, incident resolution, performance, and AI service failures. Penalty structures will be evaluated commercially and must incentivize continuous improvement throughout the contract lifecycle.

7. Technical & Operational Specifications (if applicable)

The proposed platform must conform to the following technical and operational specifications. Vendors must provide explicit compliance confirmation against each item.

7.1 Platform Architecture

- Microservices-based architecture — each business domain independently deployable and scalable.
- AI-native design — AI agent layer integrated at the platform core, not as a bolt-on module.
- Low-code/no-code configuration layer for forms, workflows, rules, fees, and AI agent behaviors.
- API-first design with documented REST/SOAP APIs for all capabilities.
- Event-driven architecture supporting real-time data propagation across domains.
- Cloud-native deployment on UAE government-approved infrastructure.
- Containerized deployment (Kubernetes or equivalent) for scalability and portability.

7.2 AI & Machine Learning Infrastructure

- Dedicated AI/ML platform with model training, versioning, deployment, and monitoring capabilities
- Support for large language models (LLMs) for document intelligence, regulatory Q&A, and conversational agent functions
- Computer vision pipeline for inspection evidence analysis — image classification, object detection, deficiency identification
- Real-time inference capability with sub-5-second response for customer-facing AI interactions
- Batch inference for nightly risk scoring updates across entire vessel and company portfolios
- Model governance framework: versioning, A/B testing, rollback capability, and performance drift detection
- All AI models trained and operated on UAE-hosted infrastructure — no personal or regulatory data to leave UAE jurisdiction for AI processing

7.3 Integration Architecture

- UAE PASS: OpenID Connect / OAuth 2.0 federated identity integration
- GISIS/IMO: RESTful API or IMO-approved protocol for flag state data exchange
- MOHRE: Secure API for seafarer employment and labor compliance data
- Emirates Classification entities: Certificate and survey record APIs
- AIS/LRIT: Real-time vessel tracking feeds with configurable refresh rates (target: sub-60-second latency)
- Global PSC databases (Paris MOU, Tokyo MOU, others): For vessel risk scoring input
- Legacy systems: ETL pipeline for complete data migration with validation

7.4 Security & Compliance

- UAE Cybersecurity Council security standards and controls framework compliance
- AES-256 encryption at rest; TLS 1.2 minimum in transit
- Role-based access control with principle of least privilege
- Multi-factor authentication for all MOEI internal and NMC users
- AI-specific security controls: protection against adversarial inputs, model poisoning, and prompt injection attacks
- Annual penetration testing by approved third party
- Full audit logging of all user actions, AI decisions, and administrative operations

7.5 Data Management

- All data and AI processing within UAE territory
- UAE Personal Data Protection Law compliance; UAE AI governance frameworks and guidelines as issued by the relevant UAE authorities
- Structured retention and archiving aligned with UAE government records standards
- Arabic and English data storage with full Unicode Arabic character set support
- Master data management ensuring consistency across all 7 domains

7.6 Infrastructure & Resilience

- Automated daily backups with tested restoration within UAE
- RPO: Maximum 4 hours; RTO: Maximum 8 hours
- Maintenance windows communicated minimum 5 working days in advance
- Annual DR test with documented outcomes provided to MOEI
- 24/7 infrastructure monitoring with automated performance alerting
- AI service resilience: AI agents must maintain degraded operation mode if AI infrastructure is partially unavailable, defaulting to human review queues

7.7 Performance Standards

- Minimum 2,000 concurrent users without performance degradation.
- Web page load: Maximum 3 seconds at 95th percentile.
- API response: Maximum 500ms at 95th percentile.
- Platform availability: Minimum 99.9% monthly uptime excluding planned maintenance windows.
- AI document verification: Maximum 30 seconds per document.
- AI inspection risk score: Maximum 60 seconds from trigger.
- AI conversational agent response: Maximum 5 seconds.
- NMC AIS data refresh: Maximum 60-second latency.

7.8 Vendor Support & Partnership

- Dedicated technical account manager and AI specialist for the contract duration
- 24/7 on-call support for P1/P2 incidents
- Structured knowledge transfer including AI model management and agent administration
- Source code escrow for platform and AI models
- Annual AI capability briefing: roadmap for new AI features, model improvements, and agentic capability expansions
- Commitment to support MOEI in achieving its AI-assisted service coverage targets throughout the contract period

8. Governance & Reporting

MOEI will establish a comprehensive governance framework covering project delivery, AI performance, platform operations, and continuous improvement throughout all project phases and into steady-state operations.

8.1 Project Governance Structure

- **Project Steering Committee (PSC)**: Chaired by MOEI senior leadership; strategic decisions, budget, scope changes, and executive escalation. Monthly during implementation, quarterly during operations.
- **Joint Project Management** Team: Co-chaired by MOEI Project Manager and Vendor Project Manager; day-to-day delivery, risk/issue management, milestone tracking. Weekly meetings.
- **AI Governance Committee**: MOEI digital leadership and vendor AI leads; responsible for AI model approvals, agent autonomy decisions, AI KPI reviews, and AI roadmap oversight. Monthly meetings.
- **Technical Architecture Board**: MOEI IT leadership and vendor technical leads; architecture decisions, integration approvals, security sign-off.
- **Change Control Board (CCB)**: Formal review of all the scope, timeline, and budget change requests.

8.2 Reporting Requirements

- **Weekly Status Report**: Delivery progress, milestones, risk/issue register, resource utilization.
- **Monthly Executive Dashboard**: Delivery KPIs, budget utilization, AI agent performance summary, open risks. Presented at Project Steering Committee meeting.
- **Monthly AI Performance Report**: Per-agent metrics — decision volume, accuracy, escalation rate, processing time, business impact. Reviewed by AI Governance Committee.
- **Quarterly Business Review:** full SLA compliance, customer satisfaction, platform usage, and AI adoption progress. Quarterly PSC.
- **Incident Report**: RCA and corrective action plan within 48 hours of P1; 5 working days for P2.
- **Annual AI Adoption Report**: AI adoption progress, model performance, and next year's AI roadmap.

8.3 Key Performance Indicators (KPIs)

Platform and AI performance KPIs to be tracked from go-live.

The below list represents the minimum set of KPIs identified at this stage — a comprehensive KPI framework covering all seven business domains will be defined and agreed jointly by MOEI and the selected vendor during the project initiation phase.

- **Platform Availability:** Minimum 99.9% monthly uptime.

- **Average Service Processing Time:** Measurable reduction per service category versus pre-platform baseline, to be established during project initiation.
- **Customer Satisfaction Score:** Minimum 90% across all service categories, measured through periodic stakeholder surveys.

- **AI Escalation Rate:** Percentage of AI agent decisions escalated to human review — tracked continuously to demonstrate improving model confidence over time.
- **AI-Assisted Service Coverage:** Progress toward MOEI's target of 80% AI-assisted service coverage — tracked and reported quarterly from go-live.

8.4 AI Governance & Control

- All new AI models and significant model updates require AI Governance Committee approval before production deployment.
- MOEI retains the right to adjust AI agent autonomy levels at any time — the platform must support real-time autonomy configuration changes without vendor involvement.
- Any AI agent found to be producing systematically biased or inaccurate output must be suspended within 4 hours of MOEI notification, pending investigation and remediation.
- An AI bias and fairness audit must be conducted periodically by an independent party, with results submitted to MOEI and the AI Governance Committee.
- MOEI retains the right to audit any AI decision and to request a full explanation of model behavior and decision logic from the vendor at any time.

8.5 Change Management & Scope Control

- All requests for changes to scope, timeline, or commercial terms must be submitted through the formal Change Control Board (CCB) process.
- All change requests must include a description, justification, and impact assessment on cost, timeline, and risk before CCB review.
- All approved Change Requests will be implemented according to a timeline and effort estimate agreed between MOEI and the vendor through the CCB, prior to implementation commencement
- MOEI reserves the right to prioritize, defer, or reject any Change Request.

8.6 Knowledge Transfer & Transition

- The vendor must provide a comprehensive Knowledge Transfer program covering both business users and MOEI technical teams, ensuring full operational independence by the end of the contract implementation phase.
- MOEI business staff trained as platform and AI agent administrators before end of implementation.

- MOEI technical staff trained in platform operations, AI model monitoring, and Tier 2 support before go-live.
- All documentation — technical, operational, user-facing, and AI governance — delivered in Arabic and English.
- Formal Operational Readiness Review including AI readiness assessment before go-live sign-off.

8 PROPOSAL CRITERIA & REQUIREMENTS

For the purpose of this RFP, the Bidder is required to submit a Technical Proposal and a Commercial Proposal. Together, these will provide a comprehensive understanding of the Bidder's capability, methodology, and cost structure.

8.4 Technical Proposal

The Technical Proposal should provide a clear and structured response demonstrating the Bidder's ability to deliver the requested solution or services in line with the RFP objectives. At a minimum, it should include:

a. **Company Profile & Credentials**
    - Legal entity details and registrations
    - Corporate profile and organizational structure
    - Relevant certifications (quality, security, compliance)

b. **Relevant Experience & References**
    - Evidence of (3) similar projects delivered successfully in the past (3 years)
    - Reference contacts for verification

c. **Solution Approach & Methodology**
    - Technical and functional approach to meeting the requirements
    - Implementation methodology and delivery timelines
    - Risk management and mitigation plan
    - Change management and training approach

d. **Capability & Resources**
    - Proposed project team, roles, and qualifications
    - Availability of resources during the project lifecycle

e. **Differentiation & Added Value**
    - Key differentiators compared to competitors
    - Innovations, efficiencies, or additional value that will benefit the Buyer

f. **Compliance Statement**
    - Explicit confirmation of compliance with the RFP's Scope of Work, Technical Specifications, and Service Level Agreements (SLAs)

8.5 Commercial Proposal

The Commercial Proposal must provide a transparent and comprehensive cost breakdown covering all aspects of implementation, operation, and ongoing support.

a. **Currency**
   - Costs to be quoted in UAE Dirhams (AED).
   - All costs to be exclusive of VAT, unless otherwise specified
b. **Payment Terms**
   - Standard terms: 90 days upon receipt of valid invoice
   - Milestone-based or phased payment models may be proposed for consideration
c. **Cost Breakdown**
   Bidders must provide a detailed breakdown covering, at a minimum:
   - **One-Time Costs** (software, hardware, integration, training, rollout, implementation services)
   - **Ongoing Costs** (subscriptions, maintenance, support, licensing, personnel)
   - **Life-Cycle Costs** (operational, consumables, upgrades, replacements)
d. **Clarity & Accountability**
   - Proposals must clearly state what is included and excluded
   - Costs not explicitly listed will not be accepted later without a formal change order
   - Non-committal wording such as "to be determined" or "details to follow" will render a proposal incomplete
e. **Justification & Benefits**
   - Explanation of cost components and rationale
   - Assessment of cost-benefit, efficiencies, or savings compared to alternatives

8.6 Buyer Rights

- The MOEI reserves the right to evaluate total cost of ownership, including operational and support costs, in addition to capital expenditure.
- The MOEI may procure hardware, software, or third-party components from alternative vendors if deemed in its best interest.
- Incomplete proposals, or those not aligned with the structure above, may be considered non-compliant.

9. SUBMISSION REQUIREMENTS

Suppliers are required to complete, sign, and stamp all forms included in **Appendix A**, and proposals must be signed by an authorized representative certifying that all information provided is true and accurate. Proposals should be prepared in a clear, structured, and comprehensive manner, directly addressing the requirements of the RFP with emphasis on completeness and clarity of content. All submissions must be provided in the format and by the deadline specified in this RFP, and be submitted in English with all financial information quoted in UAE Dirhams Currency, exclusive of VAT. By submitting a proposal, the Supplier acknowledges that it has reviewed, understood, and accepted the requirements and conditions of this RFP.

10. EVALUATION CRITERIA

Proposals will be evaluated based on, but not limited to, the following criteria:

Technical Proposal

- Supplier credentials, relevant experience, and proven ability to deliver the required scope
- Compliance with the RFP requirements, specifications, and service levels
- Clarity, quality, and completeness of the proposal in addressing all requested areas
- AI Capability & Coverage: Breadth and depth of AI enablement proposed across all seven business domains. Proposals demonstrate a credible roadmap to achieve a minimum of 80% AI-enabled service coverage.

Commercial Proposal

- Completeness, transparency, and competitiveness of the pricing structure
- Alignment with proposed contract terms and payment conditions
- Consideration of life-cycle costs, value for money, and overall cost-effectiveness